Manav Arora

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SUMMARY

4th-year AI/ML student with hands-on experience building end-to-end ML systems: from LLM-powered RAG pipelines processing SEC filings to production-scale recommendation engines and real-time detection models. Comfortable across the stack, including model training, API/backend development, and deployment.

EDUCATION

Atlas SkillTech University — **Mumbai, Maharashtra**
Bachelor of Technology in Computer Science, Artificial Intelligence & Machine Learning — **CGPA:7.7/10** | Graduation Date: Jun 2027

WORK EXPERIENCE

Plattr.io — **Remote**
AI Intern | Flask, OpenAI API, Anthropic API, FMP API — May 2025 - July 2025

- Built an end-to-end financial analysis pipeline processing SEC 10-Q filings, converting raw EDGAR data into structured datasets (18 metadata fields, 14-field exports).
- Developed 14 REST API endpoints across filings, insider trades, transcripts, and multi-mode analysis, as part of a multi-stage pipeline (query, extraction, chunking, LLM analysis).
- Integrated OpenAI and Anthropic models with chunked document processing to extract structured financial insights, including metrics, risk factors, and key company signals, from unstructured filings.

OutPulse (Open-Source) — **Mumbai, Maharashtra**
Machine Learning Engineer | YOLOv11, LiDAR — Dec 2024 - Feb 2025

- Trained a YOLOv11-based object detection model on 962 images (1,736 annotations) for cone detection, using augmentation and fine-tuning over 100 epochs.
- Built a real-time OpenCV inference pipeline from webcam and video input, achieving 0.77 mAP50 and 0.89 precision on validation data.
- Exported to ONNX for edge-compatible deployment, enabling cross-platform inference across embedded systems without cloud dependency.
- College-funded team, International Space Drone Competition (ISDC) — placed 14th/50.

Qenix Labs — **Mumbai, Maharashtra**
AI Agent Intern | FastAPI, Redis, PostgreSQL, Next.js, FFmpeg, OpenRouter — Oct 2026 - Current

- Built an end-to-end automated pipeline for 60-second vertical videos (Shorts, Reels, TikTok), taking a topic from trend research through script, video and voice generation, FFmpeg assembly with captions, automated QA checks, and a human review gate before publishing.
- Designed a Redis-backed job queue and worker service on FastAPI and PostgreSQL, with a Next.js dashboard, to run multi-stage generation jobs across 3 video models, 3 image models and S3-compatible storage.
- Engineered a multi-provider LLM layer with automatic failover across 3 models and cost-optimized model defaults, keeping the average cost of a finished Short to about $0.50.
- Developed an idea-analysis module that scores concepts on projected RPM, competition and ROI, and a competitor-video deconstruction tool that extracts a retention blueprint and generates an original script from it, across 4 content niches.

SKILLS

LLM & RAG: LangChain, FAISS, Hugging Face, OpenAI API, Anthropic API
ML/AI: NLP, Computer Vision, Recommendation Systems, PyTorch, TensorFlow, Scikit-learn
Backend: FastAPI, Flask, React
Languages: Python, SQL, C#
Databases: PostgreSQL, Redis, MongoDB, MySQL
Cloud & DevOps: Docker, Git, ONNX Runtime, Vercel, Render, Neon, Upstash
Data & Analytics: Pandas, NumPy, Matplotlib, Power BI

PROJECTS

1. **Game Recommendation System | Python, FastAPI, React, FAISS, Redis (Github) (Live Demo)** — **July 2026**
   - Designed a hybrid recommendation engine combining Bayesian Personalized Ranking (BPR), 384-dimensional FAISS embeddings, and MMR diversity reranking on 11M+ interactions across 50K+ games.
   - Optimized inference through ANN search and Redis caching, achieving 17.8% HitRate@20 with sub-50ms response times.
   - Built and deployed a full-stack application integrating Steam authentication, user library sync, and cloud-hosted APIs (Render, Vercel, Neon PostgreSQL, Upstash Redis).
2. **NLP-Based Phishing Detection System | Python, Scikit-learn, FastAPI, NLP, Browser Extension (Github)** — **March 2026**
   - Built a phishing detection system on 164K+ emails, using NLP, URL, and HTML-based features for multi-signal classification.
   - Trained a Naive Bayes model on a 20K+ token vocabulary, achieving 96.7% accuracy and 0.953 F1-score, with a low false positive rate (~1.9%).
   - Developed a real-time system with REST APIs and a browser extension, enabling sub-100ms inference and incremental retraining via feedback-driven pipelines.